

SMART LIBRARY MANAGEMENT SYSTEM

System Analysis and Desioftn Report

Course Title	System Analysis and Design
Course Code	CSE06133140

Submitted By

Sowmika Islam Suchi

Batch: 66A

Student ID: 4018

Department of Computer Science & Engineering

World University of Bangladesh

Submitted To

Md. Raihanul Haque

Lecturer

Department of Computer Science & Engineering

World University of Bangladesh

Table of Content

Topic	Page No.
Chapter 1: Planning and Requirement Analysis	1
1.1 Introduction	2
1.2 Requirement Gathering	2
1.3 Stakeholder Analysis	3
1.4 User Requirements	3
1.5 Functional Requirements	4
1.6 Non-Functional Requirements	5
1.7 Hardware Requirements	5
1.8 Software Requirements	6
1.9 Use Case Summary	6
1.10 Requirement Analysis Summary	7
Conclusion	7
Chapter 2: Feasibility Study	8
2.1 Introduction	9
2.2 Objectives of the Proposed System	10
2.3 Technical Feasibility	10
2.4 Economic Feasibility	11
2.5 Operational Feasibility	13
2.6 Schedule Feasibility	14
2.7 Legal Feasibility	15

2.8 Organizational Feasibility	15
2.9 Resource Feasibility	16
2.10 Security Feasibility	16
2.11 Advantages of the Proposed System	17
Conclusion	17
Chapter 3: System Design	18
3.1 Introduction	19
3.2 Data Flow Diagram (DFD) Concepts	19
3.3 Context Diagram (DFD Level-0)	20
3.4 DFD Level-1 (System Architecture Decomposition)	21
3.5 DFD Level-2 (Detailed Process Design)	22
3.5.1 Book Search Workflow	22
3.5.2 Book Borrowing and Returning Workflow	24
3.6 Data Dictionary	24
Conclusion	25

Chapter 1

Planning and Requirement Analysis

This chapter presents the planning and requirement analysis of the proposed Smart Library Management System for World University of Bangladesh. Planning and Requirement Analysis is a critical phase of the System Development Life Cycle (SDLC) because it helps identify the needs of users, define system requirements, and establish a clear direction for project development. The purpose of this chapter is to analyze the existing library environment, identify stakeholders, gather system requirements, and determine the resources necessary for successful implementation.

Section 1.1 provides an introduction to the planning and requirement analysis phase. Section 1.2 discusses the requirement gathering process and techniques used to collect information from stakeholders. Section 1.3 presents the stakeholder analysis, while Section 1.4 identifies the user requirements of the proposed system. Sections 1.5 and 1.6 describe the functional and non-functional requirements respectively. Sections 1.7 and 1.8 discuss the hardware and software requirements needed for implementation. Section 1.9 provides a summary of the major use cases of the system, and Section 1.10 summarizes the overall requirement analysis findings. Finally, the conclusion highlights the importance of the identified requirements and their role in guiding the design and development of the Smart Library Management System.

1.1 Introduction

Planning and Requirement Analysis is the first and one of the most important phases of the System Development Life Cycle (SDLC). The success of a software project largely depends on how accurately the requirements are identified, analyzed, and documented during this phase. The purpose of this chapter is to understand the needs of the users, analyze the existing library system, identify system requirements, and establish a clear plan for the development of the proposed Smart Library Management System.

The proposed system is intended to improve the efficiency of library operations at World University of Bangladesh by automating various activities such as book management, student monitoring, book borrowing and returning, fine calculation, and report generation. Through proper planning and requirement analysis, the project team can ensure that the developed system meets user expectations and organizational objective

1.2 Requirement Gathering

Requirement gathering is the process of collecting information from users and stakeholders to understand their needs and expectations. For the proposed Smart Library Management System, requirements were collected using several techniques.

Interview

Interviews were conducted with librarians and library staff to understand the existing workflow, operational challenges, and expectations for the new system. This method helped identify the key functions required for effective library management.

Observation

The current library activities were observed to understand how books are managed, how records are maintained, and how students interact with the library. Observation helped identify inefficiencies and limitations in the existing manual system.

Questionnaire

Feedback was collected from students through questionnaires to understand their difficulties while searching, borrowing, and returning books. Their suggestions were considered while defining the system requirements.

Document Review

Existing library records, registration forms, and borrowing documents were reviewed to understand the current data management process and reporting requirements.

1.3 Stakeholder Analysis

Stakeholders are individuals or groups who are directly or indirectly affected by the system.

Stakeholder	Role in the System
Students	Search, borrow, and return books
Librarians	Manage books and library transactions
Library Administrator	Monitor overall library activities
University Management	Review reports and system performance
CS Engineering Team	System development and maintenance

Each stakeholder has specific requirements that must be considered during system development.

1.4 User Requirements

User requirements describe the services and features expected by the users of the system.

Student Requirements

- Search books by title, author, or category.
- View book availability.
- Borrow books using student ID cards.
- Return borrowed books.
- View borrowing history.
- Check outstanding fines.

Librarian Requirements

- Add new books to the system.
- Update book information.
- Remove unavailable or damaged books.
- Issue books to students.
- Process returned books.
- Manage student records.
- Generate reports.

Administrator Requirements

- Manage user accounts.
- Configure system settings.
- Monitor system activities.
- Manage database backup and recovery.
- Generate administrative reports.

1.5 Functional Requirements

Functional requirements describe the specific functions that the system must perform.

Requirement ID	Functional Requirement
FR-01	User Login and Authentication
FR-02	Book Search
FR-03	Book Borrowing
FR-04	Book Returning
FR-05	Fine Calculation
FR-06	Book Management
FR-07	Student Management
FR-08	Report Generation
FR-09	User Account Management
FR-10	Student ID Card Verification

These functions represent the core operations of the proposed Smart Library Management System.

1.6 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards of the system.

Category	Requirement
Performance	Search results should be displayed within a few seconds
Security	Authentication and authorization mechanisms must be implemented
Reliability	Accurate and consistent data management
Usability	User-friendly and easy-to-use interface
Availability	System should be available during library operating hours
Scalability	Ability to accommodate future expansion

1.7 Hardware Requirements

The following hardware resources are required for implementing the proposed system.

Hardware	Speciflcation
Computer	Core i3/i5 Processor, 8 GB RAM
Barcode Scanner	For book identification
Student ID Scanner	For student verification
Printer	Report and document printing
Network Equipment	Internet connectivity and communication

1.8 Software Requirements

The proposed system requires the following software components

Software	Purpose
Windows/Linux	Operating System
MySQL	Database Management System
PHP/Python/Java	Application Development
HTML, CSS, JavaScript	User Interface Development
Web Browser	System Access
VS Code / IDE Code	Development and Editing
API	Version Control System
Git Postman	Testing and Debugging

1.9 Use Case Summary

The following table summarizes the major interactions between users and the system.

Actor	Activities
Student	Search books, borrow books, return books, view borrowing history
Librarian	Manage books, issue books, receive returned books, generate reports
Administrator	Manage users, monitor activities, maintain system settings

This summary provides a high-level view of how different users will interact with the proposed system.

1.10 Requirement Analysis Summary

The requirement analysis indicates that the existing manual library system faces several challenges, including time-consuming record keeping, difficulty in tracking borrowed books, errors in fine calculation, and limited access to updated information. The proposed Smart Library Management System addresses these issues by automating key library operations such as book management, borrowing and returning processes, student monitoring, fine calculation, and report generation. The analysis also identifies essential functional and non-functional requirements, along with the hardware and software resources needed to ensure smooth implementation. Overall, the requirement analysis provides a structured foundation for designing a system that meets user needs and improves the efficiency, accuracy, and reliability of library services at World University of Bangladesh.

Conclusion

Planning and Requirement Analysis provides a clear understanding of the current library environment and the expectations of stakeholders for the proposed Smart Library Management System. By identifying user requirements, functional requirements, non-functional requirements, and necessary resources, this chapter establishes a strong foundation for system design and development. The findings ensure that the proposed system will support efficient library operations, reduce manual errors, enhance user experience, and meet organizational objectives. The documented requirements will guide the next phases of the SDLC and contribute to the successful implementation of the Smart Library Management System.

Chapter 2

Feasibility Study

This chapter presents the feasibility study of the proposed Smart Library Management System for World University of Bangladesh. A feasibility study is an essential part of the System Development Life Cycle (SDLC) because it evaluates whether the proposed system is practical and achievable before moving into full-scale design and development. The main purpose of this chapter is to assess the system from different perspectives, including technical feasibility, economic feasibility, operational feasibility, schedule feasibility, and other relevant factors to ensure that the project can be implemented successfully within available resources and constraints.

2.1 Introduction

This chapter presents the feasibility study of the proposed Smart Library Management System for World University of Bangladesh.

A feasibility study is an essential part of the System Development Life Cycle (SDLC) because it evaluates whether the proposed system is practical and achievable before moving into full-scale design and development.

The main purpose of this chapter is to assess the system from different perspectives, including technical feasibility, economic feasibility, operational feasibility, schedule feasibility, and other relevant factors to ensure that the project can be implemented successfully within available resources and constraints.

2.2 Objectives of the Proposed System

The primary objective of the Smart Library Management System is to improve the overall efficiency, accuracy, and accessibility of library services at World University of Bangladesh by automating key library operations such as book catalog management, student registration and monitoring, book borrowing and returning processes, fine calculation, and report generation. The proposed system also aims to reduce manual workload, minimize record-keeping errors, enhance user experience, and provide faster decision-making support through accurate reports and real-time data availability.

2.3 Technical Feasibility

Technical feasibility refers to the process of determining whether the proposed system can be developed and implemented using the available technology, hardware, software, and technical expertise. It evaluates the technical resources required for the Smart Library Management System, including database management, application development tools, network infrastructure, and system integration to ensure smooth operation within the university library environment.

Although some hardware updates may be necessary to support devices such as barcode scanners, ID scanners, and reliable network connectivity, the proposed Smart Library Management System can be implemented using widely available technologies such as MySQL for database management, web-based programming languages, and standard computer systems. The university's existing technical resources and support staff can also contribute to the successful development, deployment, and maintenance of the system.

2.4 Economic Feasibility

Economic feasibility analyzes whether the proposed system is financially beneficial and affordable for the organization. It focuses on estimating the costs of developing, implementing, and maintaining the Smart Library Management System and comparing those costs with the expected economic benefits. The aim is to determine whether the investment is justified by the savings and efficiency improvements the system will bring to the library operations at World University of Bangladesh.

Estimated Development and Implementation Cost

Item	Quantity	Estimated Cost (BDT)
Software Development (System Design, Coding, Testing)	1 Project	80,000
Database Setup and Configuration	1	15,000
Computer Upgrade (RAM, SSD, Maintenance)	5 PCs	50,000
Barcode/Student ID Card Scanner	2 Units	10,000
Network and Internet Improvement	1 Setup	10,000
Staff Training and User Orientation	1 Session	15,000
System Documentation	1	5,000
Contingency Cost (Unexpected Expenses)	-	15,000
Total Estimated Initial Cost		200,000 BDT

Annual Operational and Maintenance Cost:

Item	Estimated Annual Cost (BDT)
System Maintenance and Updates	20,000
Hardware Maintenance	10,000
Database Backup and Security	5,000
Technical Support	15,000
Total Annual Cost	50,000 BDT

Expected Economic Benefits

Benefit Source	Estimated Savings (BDT/Year)
Reduced paper, printing, and stationery costs	20,000
Reduced manual record management workload	25,000
Reduced book loss and misplacement	20,000
Improved operational efficiency and time savings	15,000
Reduced report preparation and administrative costs	10,000
Total Estimated Annual Benefit	90,000 BDT

Cost-Benefit Analysis

- Initial Development Cost = **200,000 BDT**
- Annual Maintenance Cost = **50,000 BDT**
- Annual Benefit = **90,000 BDT**

Net Annual Benefit = Annual Benefit - Annual Maintenance Cost

= 90,000 – 50,000

= **40,000 BDT per year**

It is estimated that the university can save approximately 60,000-80,000 BDT per year through improved efficiency, reduced paperwork, fewer errors, and better control over library resources. Over time, these savings can offset the initial development cost, making the system economically viable for the university library.

Therefore, considering the moderate implementation cost and the significant long-term benefits, the Smart Library Management System is economically feasible. The investment is justified because it will reduce operational costs, improve service quality, and support better management decisions through accurate and timely reports

2.5 Operational Feasibility

Operational feasibility determines whether the proposed system will function properly within the existing organizational environment and whether the users (librarians, students, and administrators) will accept and effectively use the Smart Library Management System. It focuses on evaluating the readiness of the university library to adopt the new system, the changes required in day-to-day operations, the training needs of staff, and the overall impact of the system on library services. Since the proposed system is designed to simplify book management, borrowing/returning transactions, fine calculation, and report generation, it is expected to enhance operational efficiency and reduce the workload of library personnel while providing better service to students.

Since the proposed system will improve efficiency and service quality, it is likely to be accepted by the library staff and students. The system will reduce manual record keeping, minimize errors, and ensure faster access to accurate information. Although some training and adjustment time may be required for librarians and staff to become familiar with the new system, the user-friendly interface and automation features will make the transition smoother. Therefore, the Smart Library Management System is operationally feasible because it can be integrated into the existing library workflow and will support improved library operations.

2.6 Schedule Feasibility

Schedule feasibility refers to the analysis of whether the proposed project can be completed within the required time frame using the available resources. It evaluates the estimated time needed for requirement analysis, system design, development, testing, implementation, and user training to ensure that the Smart Library Management System can be delivered on time without unnecessary delays. A realistic schedule helps ensure proper planning, effective resource utilization, and timely deployment of the system for World University of Bangladesh library operations.

Activity	Duration	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Requirement Analysis	2 Weeks						
System Design	2 Weeks						
System Development	8 Weeks						
Testing	2 Weeks						
Implementation	1 Week						
User Training	1 Week						

2.7 Legal Feasibility

Legal feasibility examines whether the proposed system follows legal requirements, policies, and regulations related to software usage, data privacy, intellectual property, and institutional rules. For the Smart Library Management System, it is important to ensure that the system properly protects student and staff information, maintains confidentiality of borrowing records, and uses licensed software tools and technologies where required. The system should also follow university guidelines for data retention and access control so that only authorized users can view or modify sensitive information.

Therefore, the Smart Library Management System is legally feasible.

2.8 Organizational Feasibility

Organizational feasibility determines whether the proposed system supports the goals, structure, and overall direction of the organization and whether it can be adopted successfully within existing management and operational practices. The Smart Library Management System aligns with the objective of World University of Bangladesh to improve service quality, increase operational efficiency, and adopt modern digital solutions for managing academic resources. By enabling faster book searching, accurate record keeping, automated fine calculation, and reliable reporting, the system will strengthen library services and help the university achieve improved resource utilization and better decision-making through real-time information. With proper training and management support, the organization will be able to implement and sustain the system effectively.

Therefore, the Smart Library Management System is organizationally feasible.

2.9 Resource Feasibility

Resource feasibility analyzes whether the required resources are available for successful implementation of the project. The university already has basic resources such as computers, internet facilities, and technical support staff. However, some computers and equipment require upgrading because many devices are currently slow and outdated. Additional resources such as barcode scanners or ID card scanning systems may also be required for proper implementation. Although the library space is limited, the proposed digital system will help reduce paperwork and improve space management. Since most required resources are already available and additional improvements can be implemented gradually, the project is manageable within the university environment.

Therefore, the Smart Library Management System is resource feasible.

2.10 Security Feasibility

Security feasibility evaluates whether the proposed system can ensure proper security for data, users, and system operations against unauthorized access, misuse, and threats. For the Smart Library Management System, security is important because it will store sensitive information such as student profiles, borrowing history, fines, and administrative records. The system should implement role-based access control so that students, librarians, and administrators can access only the functions relevant to their roles. Strong authentication methods, secure password storage, session management, and audit logs will help prevent unauthorized activities and maintain accountability. Additionally, database security measures such as restricted permissions and secure configuration will reduce the risk of data leakage or manipulation.

Regular backups and proper monitoring systems are also necessary to protect against data loss due to system failure, accidental deletion, or cyber incidents. The system can be deployed using secure server practices, updated software components, and encrypted communication (where applicable) to ensure safe access through the network. By combining access control, monitoring, backups, and secure database management, the proposed system can maintain confidentiality, integrity, and availability of library data.

2.11 Advantages of the Proposed System

The Smart Library Management System will provide several important advantages for World University of Bangladesh by improving the overall quality and efficiency of library services. It will reduce manual workload and paperwork by automating book cataloging, borrowing/returning transactions, fine calculation, and report generation. Students will benefit from faster searching, real-time availability checking, and improved access to accurate information, while librarians will be able to manage books and member records more efficiently with fewer errors. The system will also enhance monitoring and control over library resources, reduce book misplacement and loss, and provide reliable reports to support better decision-making for administrators and university management.

Conclusion

After analyzing all aspects of feasibility—technical, economic, operational, schedule, legal, organizational, resource, and security—it is clear that the proposed Smart Library Management System is practical and achievable for World University of Bangladesh. The system can be developed using available technologies, implemented within a realistic schedule, and supported through manageable costs when compared with the expected benefits. With appropriate planning, training, and management support, the system can be integrated into existing library operations successfully.

The proposed system will modernize the traditional library system by providing automation, accuracy, improved security, and better service delivery. It will support efficient management of books and users, reduce errors in record keeping and fine calculation, and provide timely reports for decision-making. Overall, the feasibility study indicates that implementing the Smart Library Management System will be beneficial for the university and will contribute to improved library performance and user satisfaction.

Chapter 3

System Design

This chapter presents the system design of the proposed Smart Library Management System for World University of Bangladesh. System design is a vital phase of the System Development Life Cycle (SDLC) because it transforms the requirements identified in the previous chapter into a structured blueprint that guides system development. The purpose of this chapter is to describe how the system will be organized, how data will flow between users and processes, and how the major components of the system will interact to support efficient library operations.

3.1 Introduction

The purpose of the system design phase is to provide a clear blueprint of how the proposed Smart Library Management System will be developed and implemented. This phase focuses on defining the system architecture, identifying the major processes, designing the database structure, and describing the interactions between users and the system. A well-planned design ensures that the developed system meets all functional and non-functional requirements efficiently and reduces the possibility of errors during implementation.

The proposed system has been designed to support student ID card verification, efficient book searching, borrowing and returning workflows, fine calculation, and report generation. The design also ensures secure access through role-based authentication for students, librarians, and administrators. By modeling the system using diagrams and structured descriptions, the design provides a complete and understandable representation of how the system will operate in the university library environment.

3.2 Data Flow Diagram (DFD) Concepts

A Data Flow Diagram (DFD) is a graphical representation of how data moves through a system and how different processes transform that data. It helps visualize the flow of information between external entities (users), system processes, and data stores. DFDs are widely used during system design to ensure that system requirements are correctly represented and that all major interactions are clearly understood before implementation.

The Smart Library Management System uses standard DFD symbols to represent the system components and the movement of data. These diagrams illustrate how students, librarians, and administrators interact with the system for activities such as searching books, borrowing and returning, fine calculation, and report generation. DFDs also help identify the required data stores and ensure that the system processes are logically organized and connected.

External Entity (Rectangle): Represents external users or organizations that interact with the system, such as Students, Librarians, Administrators, and CS Engineering.

Process (Rounded Rectangle): Represents activities performed by the system, such as student verification, book searching, borrowing books, returning books, fine calculation, and report generation.

Data Store (Open Rectangle): Represents databases where information is permanently stored, including Student Database, Book Database, Circulation Database, Fine Database, and System Log Database.

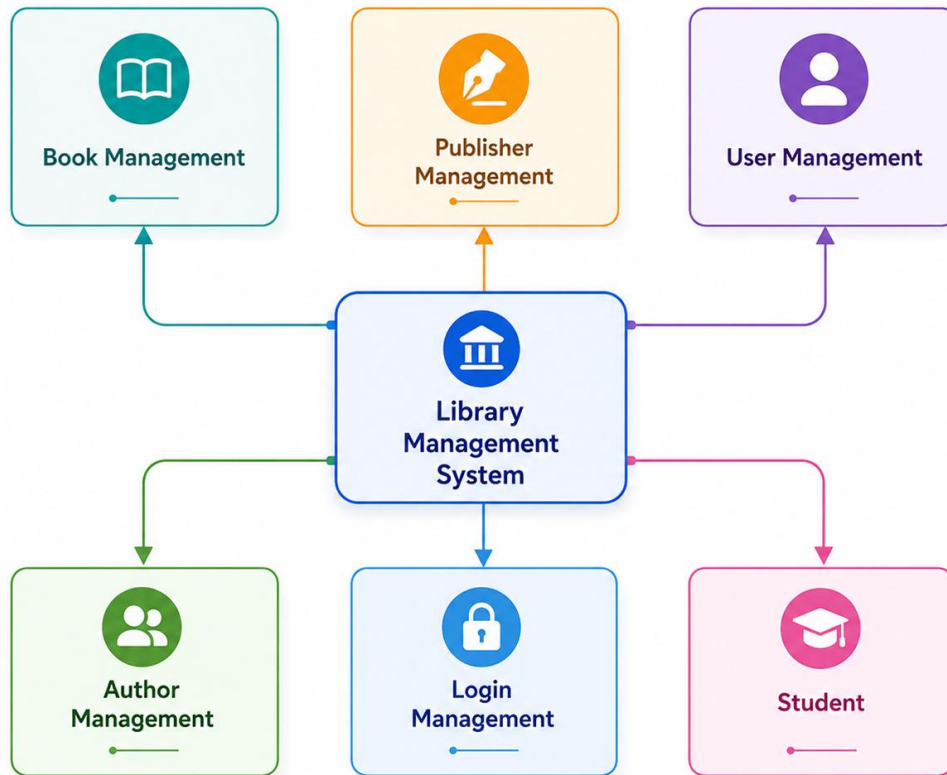
Data Flow (Arrow): Represents the movement of data between external entities, system processes, and data stores.

3.3 Context Diagram (DFD Level-0)

The Context Diagram, also known as the Level-0 Data Flow Diagram, provides the highest-level overview of the Smart Library Management System. In this diagram, the entire application is represented as a single process that communicates with external entities. It defines the overall scope of the system and illustrates how users exchange information with the proposed system.

Students interact with the system to search books, borrow books using their student ID cards, return borrowed books, and receive borrowing confirmations, book availability information, and fine notifications. Librarians manage book circulation, maintain library records, update book information, and receive circulation reports. Administrators manage user accounts, configure system settings, monitor system activities, and generate administrative reports. CS Engineering provides technical support, maintenance, and future software updates for the system.

DFD Level-0: Context Diagram :



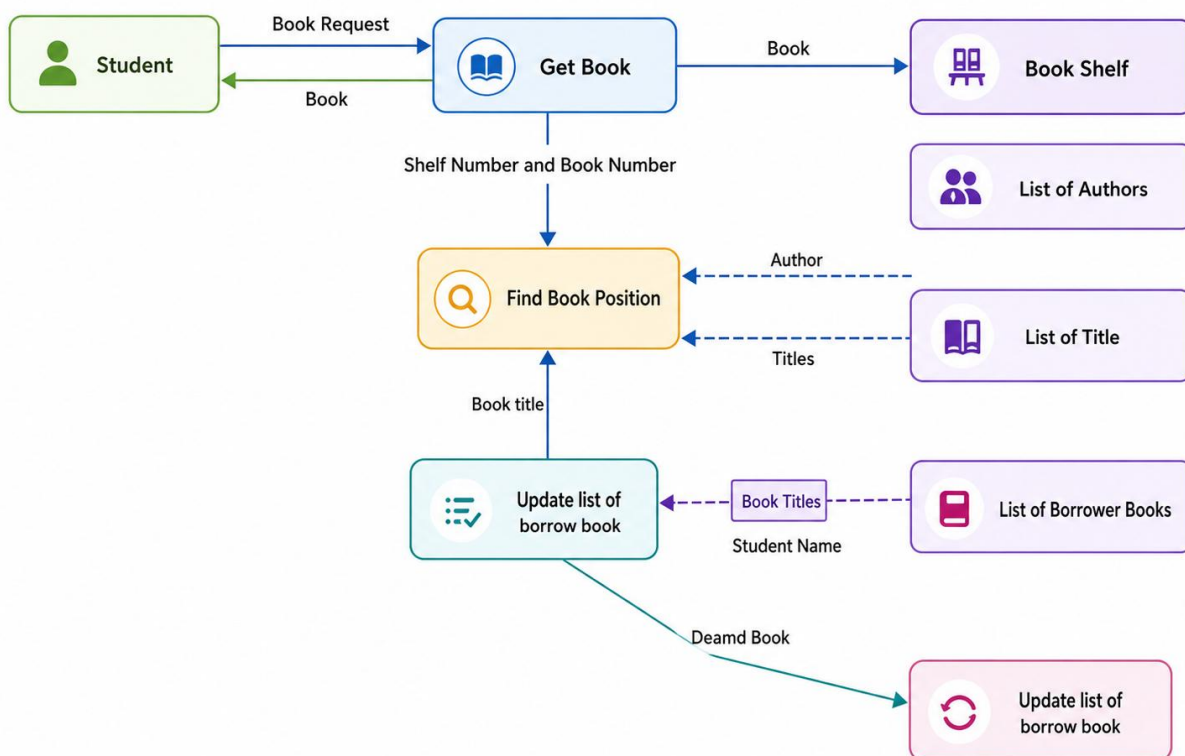
3.4 DFD Level-1 (System Architecture Decomposition)

The Level-1 Data Flow Diagram divides the Smart Library Management System into major processes to show how different functional components work together. Unlike the context diagram, the Level-1 DFD breaks the system into multiple processes and illustrates how data moves between those processes, external entities, and data stores. This decomposition provides a clearer understanding of the system architecture and ensures that each key function is represented in the overall system design.

The first process is Authentication and Student Verification, where users provide login credentials or student ID information to access the system. Once verified, users can proceed to other processes such as Book Search, Book Borrowing, Book Returning, Fine Calculation, and Report Generation. Each process receives inputs from users or other processes, performs system operations, and generates outputs such as search results, borrowing confirmations, return status, fine amounts, and printable or downloadable reports for management purposes.

The system uses several data stores including the Student Database, Book Database, Circulation/Transaction Database, Fine Database, and System Log Database. The Student Database maintains member profiles and verification details, while the Book Database stores catalog information and availability status. Borrowing and returning activities are recorded in the Circulation Database, fines are stored and updated in the Fine Database, and all significant actions are tracked in the System Log Database for monitoring and security. This Level-1 DFD ensures that data is properly organized and flows logically between processes and storage components.

DFD Level-1: System Architecture



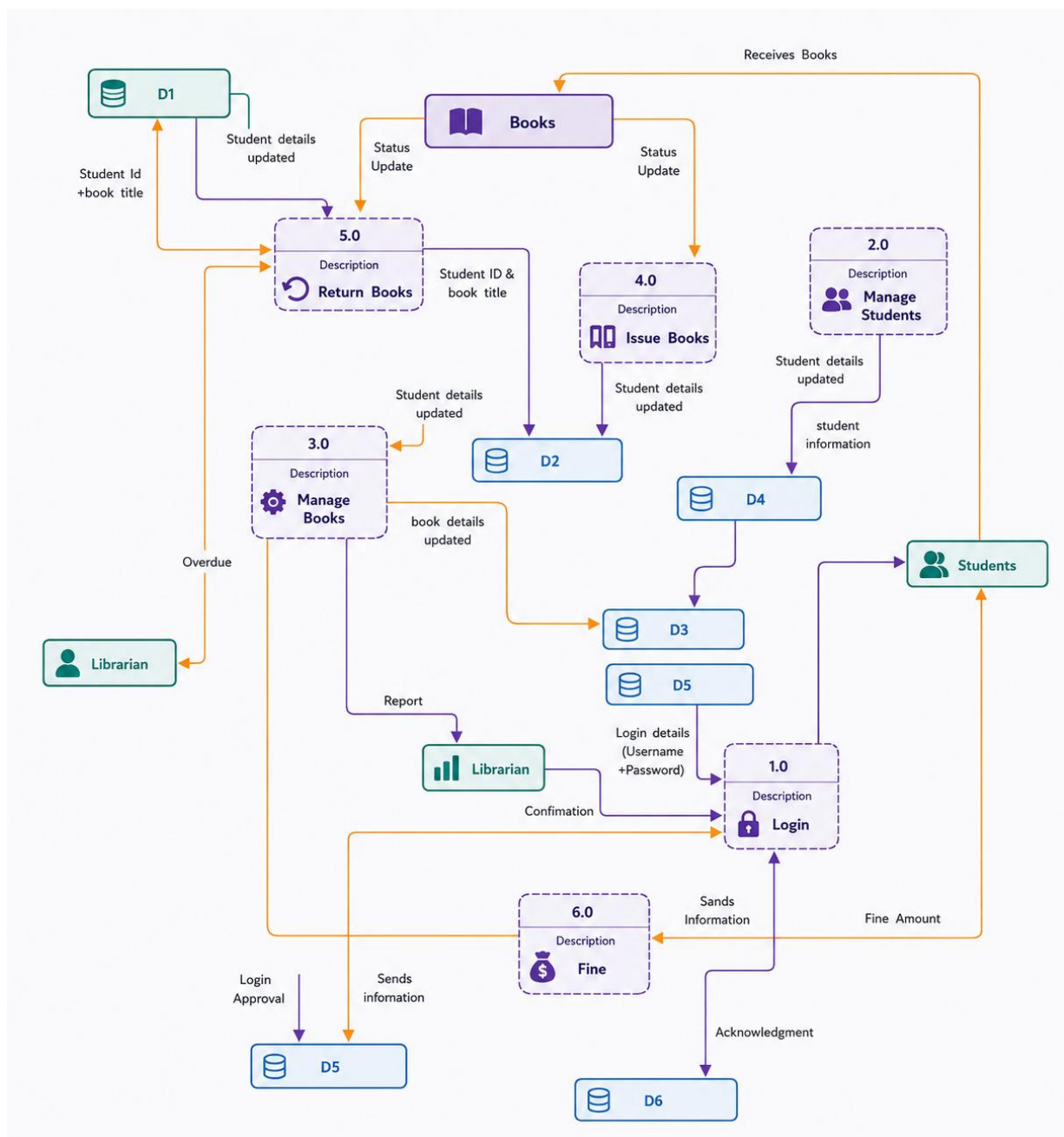
3.5 DFD Level-2 (Detailed Process Design)

The Level-2 Data Flow Diagram provides a more detailed view of the major processes presented in the Level-1 DFD by breaking them down into smaller sub-processes. This level focuses on how specific tasks are performed within each main process, the sequence of activities, and the exact data flows between sub-processes and data stores. The Level-2 DFD helps ensure that the system logic is complete and that every required operation is clearly defined for implementation.

3.5.1 Book Search Workflow

The Book Search process begins when a student enters a search request using keywords such as book title, author name, ISBN, or category. The system receives the search query and validates the input, then sends a request to the Book Database to retrieve matching records. After processing the query, the system displays the search results along with relevant details such as book title, author, category, shelf location, and availability status. If the book is available, the student can proceed to request borrowing through the borrowing process; if unavailable, the system informs the student and may show alternative available books or related suggestions depending on system configuration.

DFD Level-2: Book Search Process



3.5.2 Book Borrowing and Returning Workflow

The borrowing process starts when the librarian scans the student's ID card. The system verifies the student's membership status from the Student Database. If the student is eligible, the librarian scans the book barcode. The system checks the availability of the selected book. If the book is available, a borrowing transaction is recorded in the Circulation Database, and the book status is updated to 'Issued'. Finally, a borrowing receipt is generated.

During the return process, the librarian scans the returned book barcode. The system retrieves the corresponding borrowing record and compares the return date with the due date. If the book is returned after the due date, the Fine Calculation Module computes the overdue fine and stores it in the Fine Database. If the book is returned on time, the loan record is updated and the book status is changed back to 'Available'.

3.6 Data Dictionary

The Data Dictionary defines the database structure used in the Smart Library Management System. It specifies each table, its attributes, data types, constraints, and relationships between tables. The data dictionary ensures consistency during database implementation and simplifies future maintenance.

The Student table stores student information including Student ID, full name, department, and borrowing status. The Book table contains information about each book such as ISBN, title, author, category, shelf number, and availability status. The Circulation table records book borrowing and return transactions including issue date, due date, and return date.

Additional tables such as the Fine Table, Category Table, and System Log Table are used to maintain overdue fines, book classifications, and administrative activities respectively.

Conclusion

The system design presented in this chapter provides a complete logical and structural framework for the Smart Library Management System. The Context Diagram, Level-1 DFD, Level-2 DFD, and Data Dictionary collectively describe how data flows through the system, how different modules interact with one another, and how information is organized within the database. The proposed design supports efficient book management, secure student verification, automated fine calculation, accurate reporting, and effective library administration. Therefore, the system design serves as a strong foundation for the implementation and future development of the Smart Library Management System.